

BloodHound Attack Path Evidence Report

Internal Network & Active Directory Pentest Evidence Documentation

Field	Details
Test Area	BloodHound attack path
Evidence Source	15-bloodhound-attack-path-domain-admin.png
Result Type	Attack path visualized
Primary Tool / Component	BloodHound / Neo4j
MITRE ATT&CK Mapping	T1069.002 - Domain Groups; T1482 - Domain Trust Discovery
Document Purpose	Professional GitHub-ready evidence report with redacted screenshot and interpretation.

This report documents one specific evidence row from the AD Network assessment. It is intentionally evidence-specific, not a generic attack tutorial. Sensitive values have been blurred or represented as redacted placeholders for safe portfolio publication.

1. Embedded Evidence Screenshot

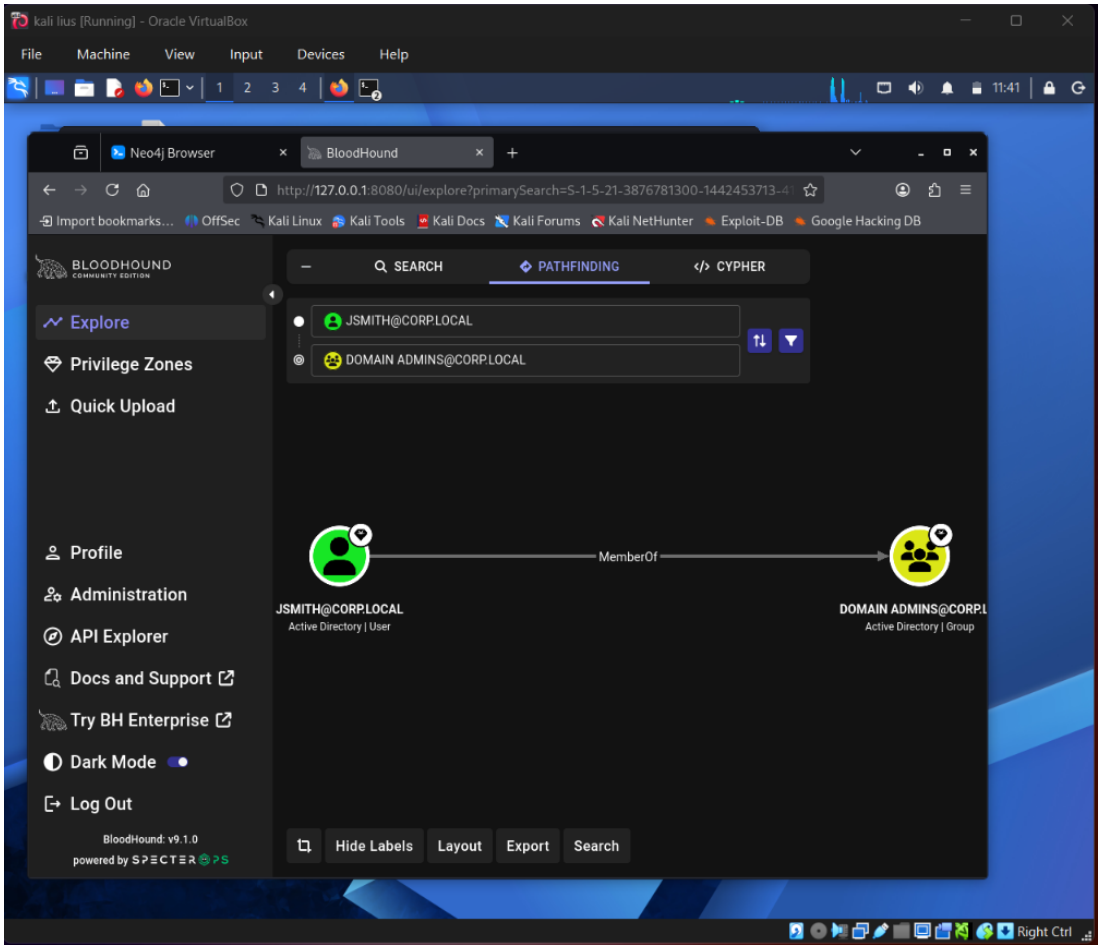


Figure 1 - Redacted evidence screenshot: 15-bloodhound-attack-path-domain-admin.png

What this screenshot proves

The screenshot shows BloodHound visualizing a path from an owned or tested principal toward a high-value target such as Domain Admins.

Evidence Context

Area	Value
Testing Phase	Active Directory Attack Path Analysis
Tool / Component	BloodHound / Neo4j
Objective	Visualize relationships between users, groups, computers, and privileges to identify paths toward domain-level control.
Result Type	Attack path visualized

2. Evidence-Specific Methodology

This section explains how this evidence fits into the overall internal network and Active Directory attack path. The goal is to demonstrate professional testing logic, not simply show a terminal screenshot.

- Collect domain relationship data using SharpHound/BloodHound collector.
- Import collected data into BloodHound.
- Mark validated credentials or accounts as owned.
- Run path queries such as shortest path to Domain Admins.

Command / Workflow Used

```
bloodhound-python -d <DOMAIN> -u <USER> -p <REDACTED_PASSWORD> -dc <DC_IP> -c all --zip  
# BloodHound GUI: import zip -> run path analysis queries
```

Sanitized Output Style

Shortest path to high-value target identified in BloodHound graph. Specific node labels may represent lab accounts and groups.

Assessment Interpretation

The evidence supports the result type: Attack path visualized. In a professional assessment, this evidence would be paired with the exact testing scope, timestamps, target identifiers, and remediation ownership. For GitHub publication, the report intentionally avoids showing live secrets, passwords, hashes, or reusable credential material.

3. Risk Analysis

Misconfigured group memberships, local admin rights, or delegated privileges can create unexpected escalation paths that are difficult to see without graph analysis.

Why this matters in enterprise AD environments

- Active Directory attacks often chain multiple medium-risk issues into domain compromise.
- Credential material, even when not plaintext, can become dangerous if cracking, relaying, or alternate authentication is possible.
- Graph-based permissions and relationship analysis is necessary because AD risk is often hidden in group nesting and delegated rights.
- Evidence must be documented clearly so defenders can connect the technical finding to a specific control gap.

Recommended Remediation

- Review privileged group membership regularly.
- Remove unnecessary local admin rights.
- Use tiered administration.
- Audit delegated permissions.
- Run BloodHound defensively and remediate shortest paths to high-value targets.

Detection and Monitoring Opportunities

- Correlate authentication events with unusual source hosts and off-hours activity.
- Monitor SMB, LDAP, Kerberos, and replication-related event patterns.
- Alert on abnormal use of administrative protocols and credential dumping indicators.
- Review domain privilege assignments and privileged group membership changes regularly.

4. GitHub Publication and Redaction Guidance

This report is prepared for safe GitHub portfolio publication. The embedded screenshot has been treated as evidence, but sensitive material must not be recoverable or readable.

Sensitive Item	Action for GitHub
NTLM / NetNTLMv2 hashes	Blur or remove completely. Never publish reusable hash material.
Kerberos TGS / krb5tgs values	Blur or replace with <REDACTED_TICKET>.
Plaintext passwords	Do not publish. Replace with <REDACTED_PASSWORD>.
DCSync / NTDS hashes	Treat as domain compromise material and fully redact.
Internal IPs / domain names	May remain for lab context, but redact if privacy is required.
Username / service accounts	Keep only if lab-only; otherwise redact or generalize.

Suggested GitHub Link

[Open Report](./reports/evidence/06-bloodhound-attack-path-evidence-report.pdf)

Professional Note

This document should be stored with the AD Network module reports, preferably under 03-ad-network/reports/evidence/. It is designed to replace a plain screenshot filename in the Evidence-Based Testing Summary table with a professional clickable report.